3D BHARAT

“A Revolution of Visualization”

1. Introduction

3D Bharat is a unified digital platform developed for the planning, design, measurement, execution, and monitoring of construction and infrastructure projects. It integrates 3D modeling, field data collection, and drone-based imagery to provide a complete and accurate representation of project progress. It also incorporates **Digital Twin technology along with IoT, GNSS RTX, and GPS-based** solutions to enable real-time data synchronization, high-precision positioning, and intelligent project monitoring.

Unlike traditional construction planning tools, 3D Bharat enables users to visualize, design, and measure infrastructure in a highly interactive and simplified manner, making it accessible to both technical and non-technical users.





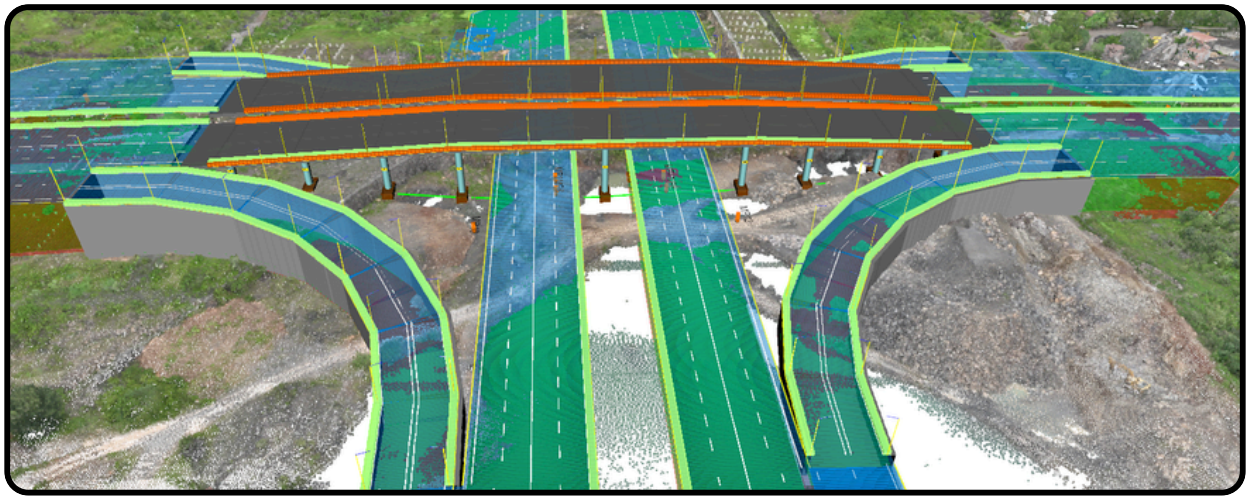
2. Vision & Objectives

Vision

To revolutionize infrastructure development by providing a simple, accurate, and digital-first platform for all types of infra development projects.

Objectives

- To simplify construction **design and planning** using 3D visualization
- To provide accurate quantity estimation **(cutting, filling, material usage)**
- To enable real-time project monitoring using drone and field data
- To reduce dependency on complex market available tools
- To integrate multiple infrastructure domains into a single platform



Interchanging Point

3. Key Modules of 3D Bharat

3D Bharat consists of an Main Modules

3.1 3D Bharat Web Module

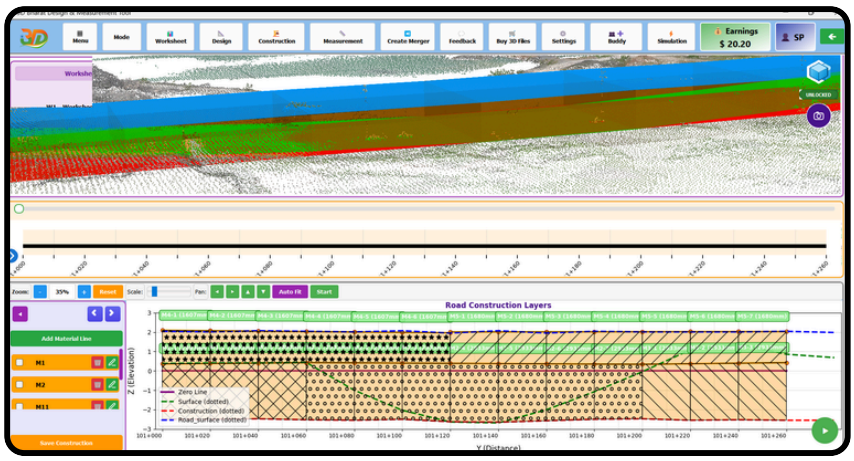
- The 3D Bharat Web Module is one of the most important components of the platform. It is used to configure construction projects by defining the **departments, contractors**, their **authorities**, and **work responsibilities.**
- The platform consists of four main roles:
  - Platform Owner
  - Department
  - Contractor
  - Administrator
- Both the Department and the Contractor can create and manage construction projects.
- The module also allows drone operation management by assigning contractors responsible for drone surveys. The assigned contractor can upload drone data to the server against the respective project and KM (chainage).
- Additionally, the module provides **role-based access** control, enabling **administrators** to assign permissions and responsibilities to employees based on their designated work areas.

3.2 3D Bharat Design Module

In Design Module we are designed multiple type construction 3D Desifn like Road, Bridge, Tunnel, Metro and Railways.

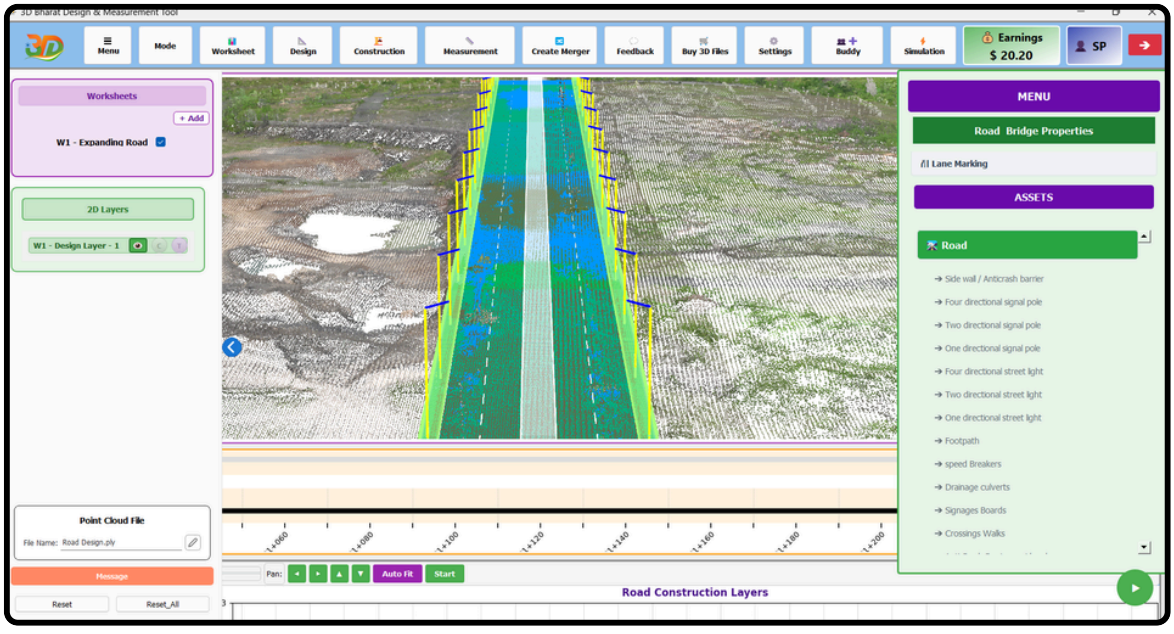
3.2.1 Road Design

This module allows users to design and analyze road construction projects.



Features:

- Road alignment and 3D design
- Cutting and excavation quantity calculation.
- Layer-wise material filling
- BOM **(Bill of Materials)** generation
- Asset mapping including:
  - Street lights
  - Signal poles
  - Dividers
  - Side walls
  - Signage boards



3.2.2 Bridge Design

This module focuses on bridge structure modeling and material estimation.

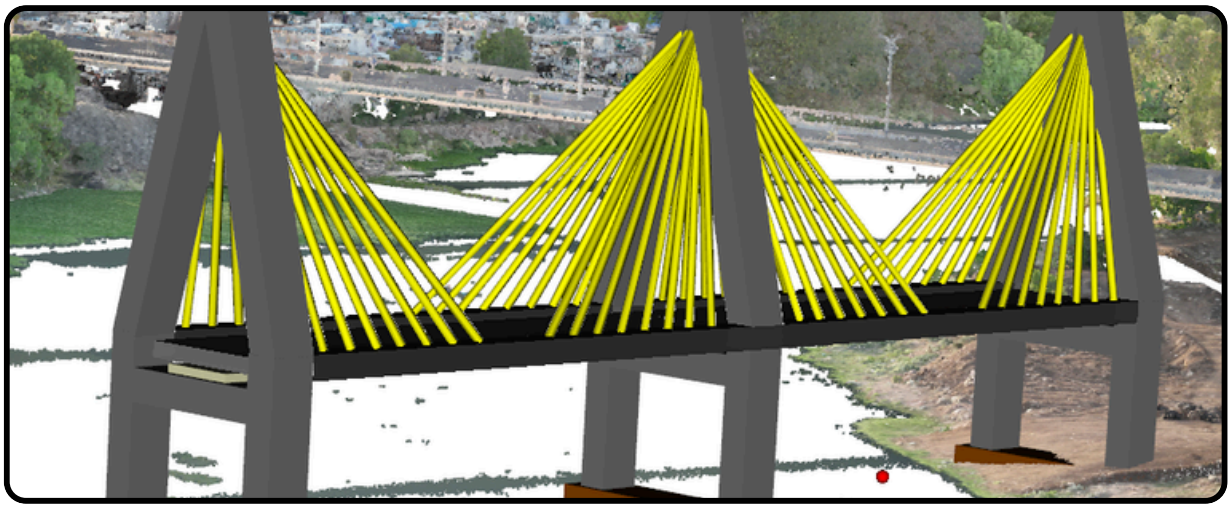
Features:

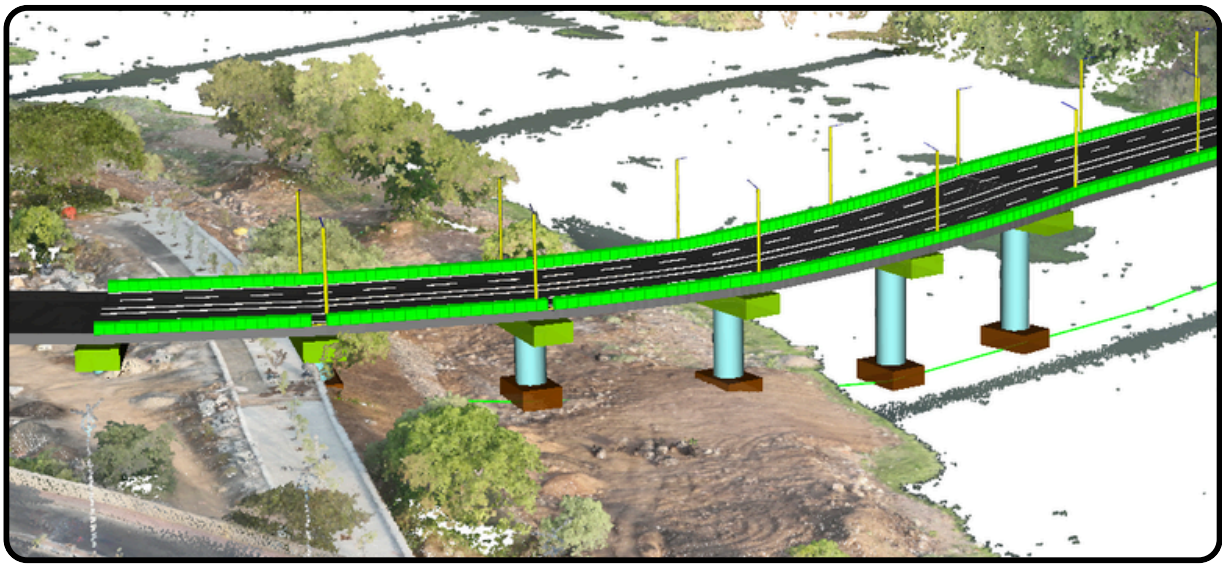
- Design bridge components with configurable dimensions.
- Configure **Footing** and **Pile** Foundations.
- Design **reinforcement bars** with grade and quantity.
- Estimate concrete and steel quantities automatically.
- Generate **Bill of Materials (BOM).**
- Configure **pillars, pier caps, girders, and deck slabs.**
- Support multiple span bridge designs.
- Configure **cable stay towers and cables.**
- KM and chainage-based bridge positioning.
- Export design and quantity reports.

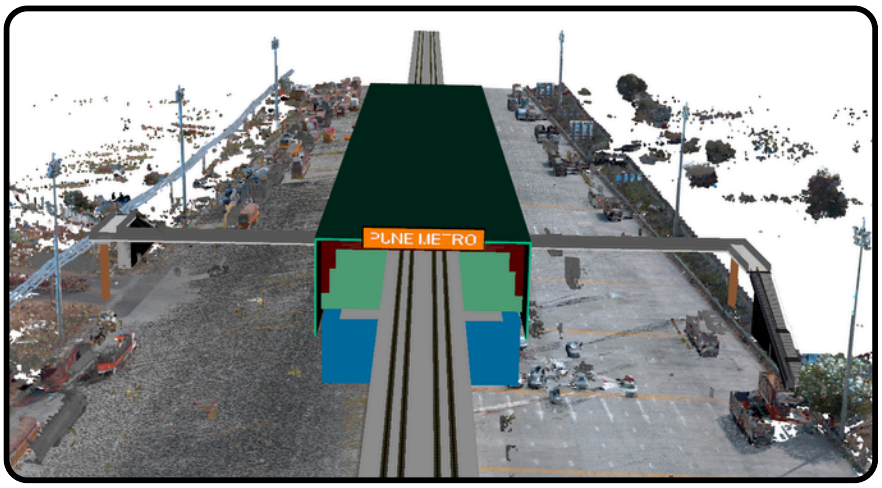


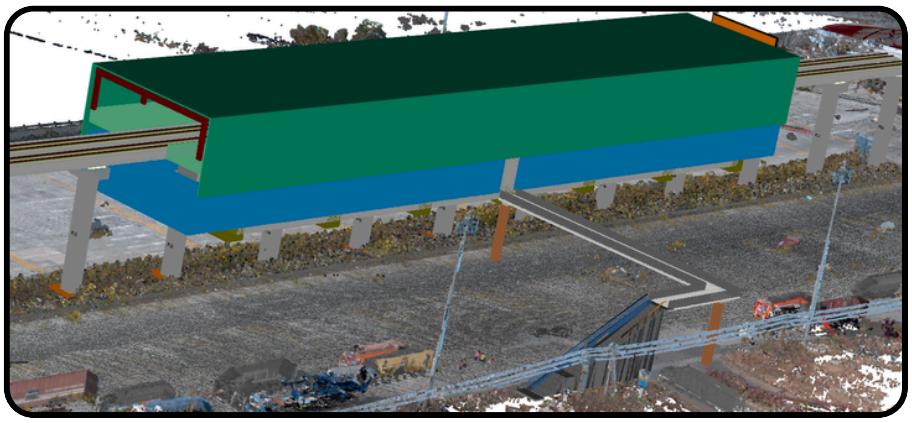
Supported Bridge Types:

- Single Deck Bridge
- Parallel Deck Bridge
- A-Shape Cable Stay Bridge
- H-Shape Cable Stay Bridge









3.2.3 Metro Design

This module is used for metro infrastructure modeling.

Features:

- Design of metro pillars:
  - **U-shape**
  - **L-shape (cantilever)**
  - Standard vertical pillars
- Metro girder box design
- **Structural layout** visualization
- Metro Station
- Metro Yards
- Metro Assets:
  - OHE Pole
  - Singal Pole
  - Signage Board
  - Footover Bridge
  - Staircase
  - Elevator
  - Metro Light Pole
  - Anti Crash Barrier

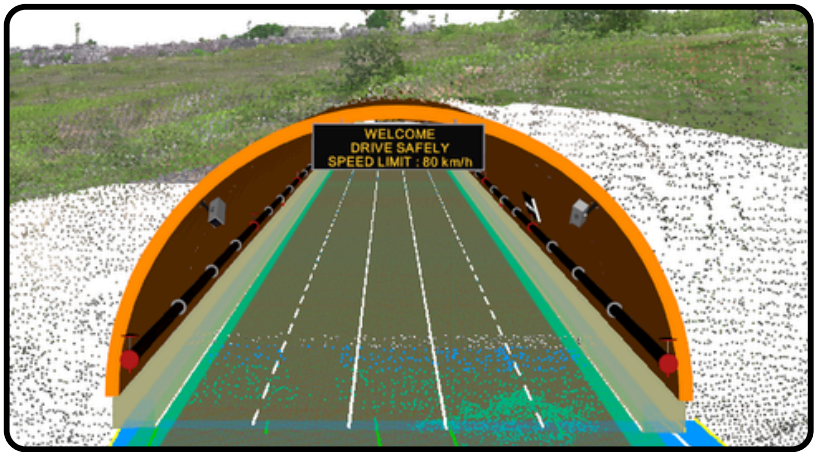
3.2.4 Tunnel Design

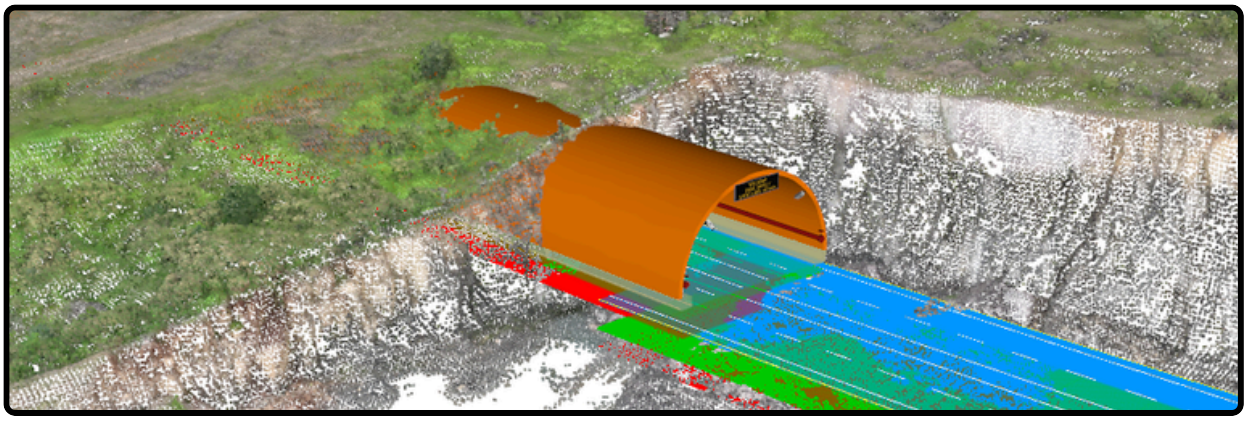
The Tunnel Design module enables users to **design tunnels using arc points** and road design.

In 3D Bharat, tunnels are designed in alignment with the road design and defined using arc points. The tunnel design process involves the following steps:

- Mark the polygon.
- Cut the point cloud surface.
- Define the tunnel arc points and specify the wall thickness.
- Design the road using the baseline.
- Map the lane markings.
- Map tunnel assets with reference to the arc length and corresponding arc points.

This workflow ensures accurate tunnel modeling and seamless integration with the road design while maintaining precise asset positioning and alignment.





3.2.5 Pipeline Design

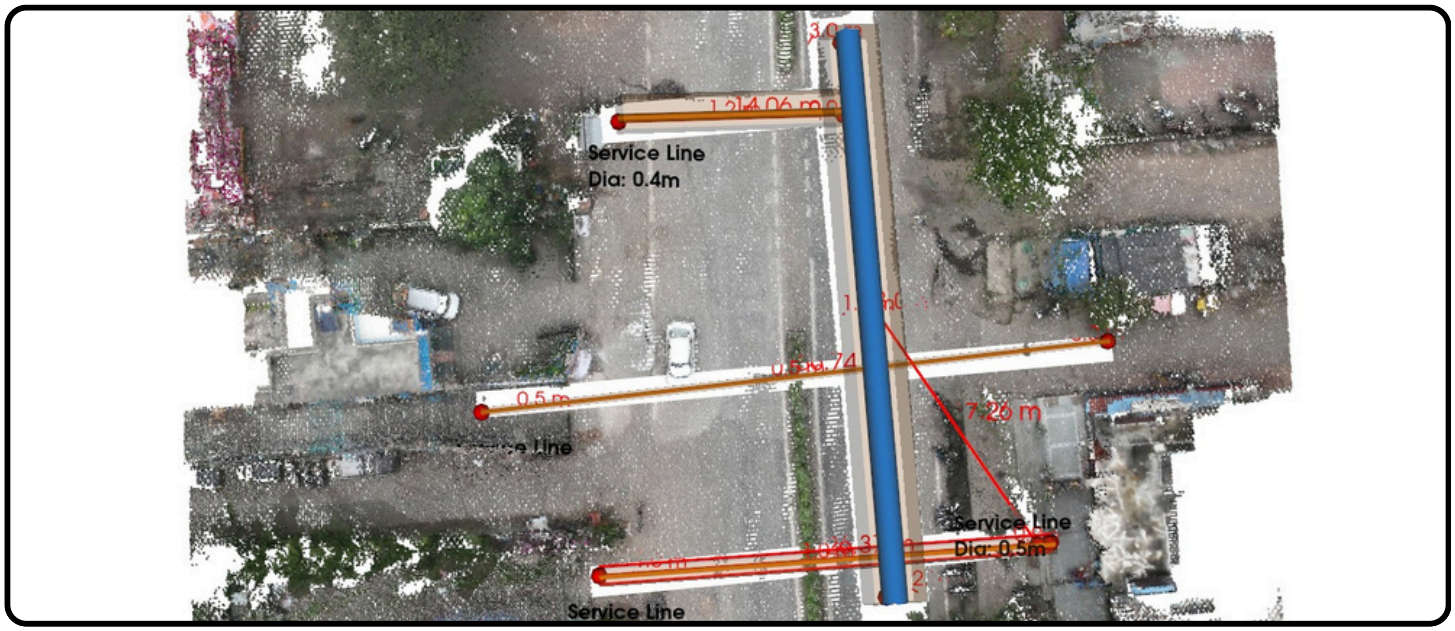
The Pipeline Design module in 3D Bharat enables the planning, design, and management of **underground utility networks** within infrastructure projects. It provides an accurate 3D representation of **buried utilities, making future maintenance and expansion** more efficient.

Features

- Design and visualize underground pipeline networks in a 3D environment.
- Configure multiple utility pipelines, including:
  - Water supply pipelines
  - Gas pipelines
  - Sewerage pipelines
  - Stormwater drainage pipelines
  - Electrical conduit pipelines
  - Telecommunication/optical fiber ducts
- Define pipeline alignment, depth, diameter, and material specifications.
- Integrate underground pipelines with road, bridge, metro, and urban infrastructure designs.
- Map pipelines with **precise geospatial coordinates** and reference points.
- Visualize the complete underground utility network for better planning and execution.

The Pipeline Design Module ensures that all underground utilities are digitally mapped and accurately documented, providing a reliable foundation for future construction, maintenance, and urban development projects.



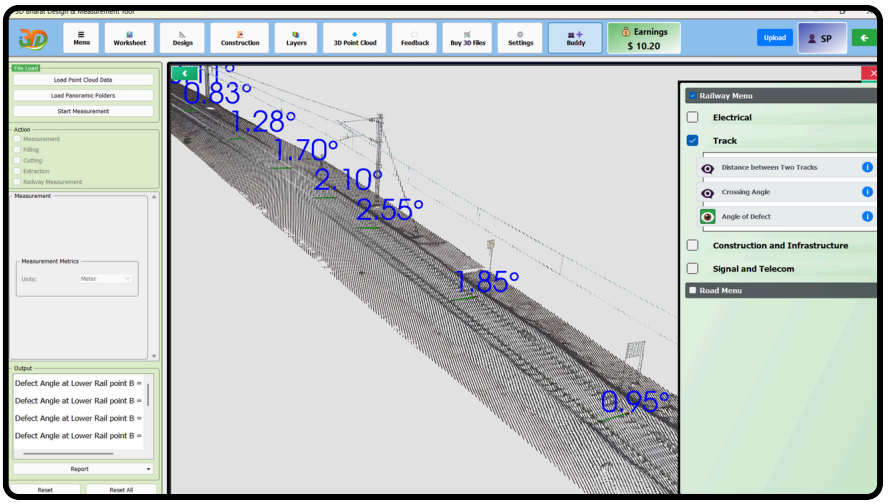


5.3D Bharat Measurement Module

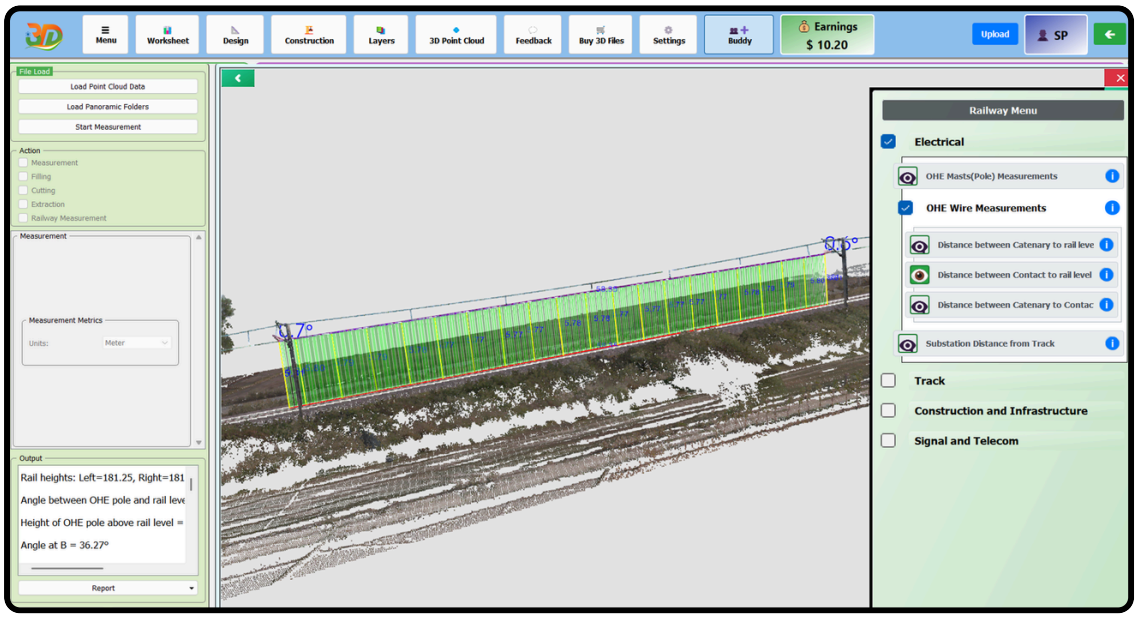
Measurement is an integrated capability available across all design modules within 3D Bharat.

Capabilities:

- **Measurement of:**
  - Height
  - Width
  - Angles
  - Surface area
  - Volume
- **Applicable to**:
  - Road designs
  - Bridge structures
  - Metro components
  - Railway assets
  - Buddies shared designs & worksheets



- With this measurement module we can do precise work progress monitoring against the design data.



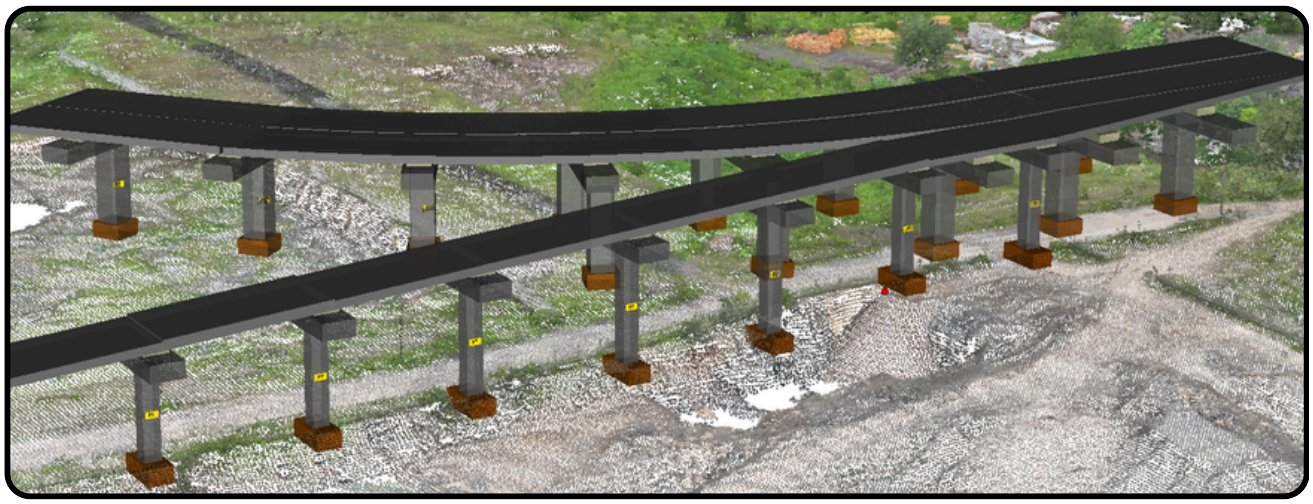


6. Design Merging & Intersection Management

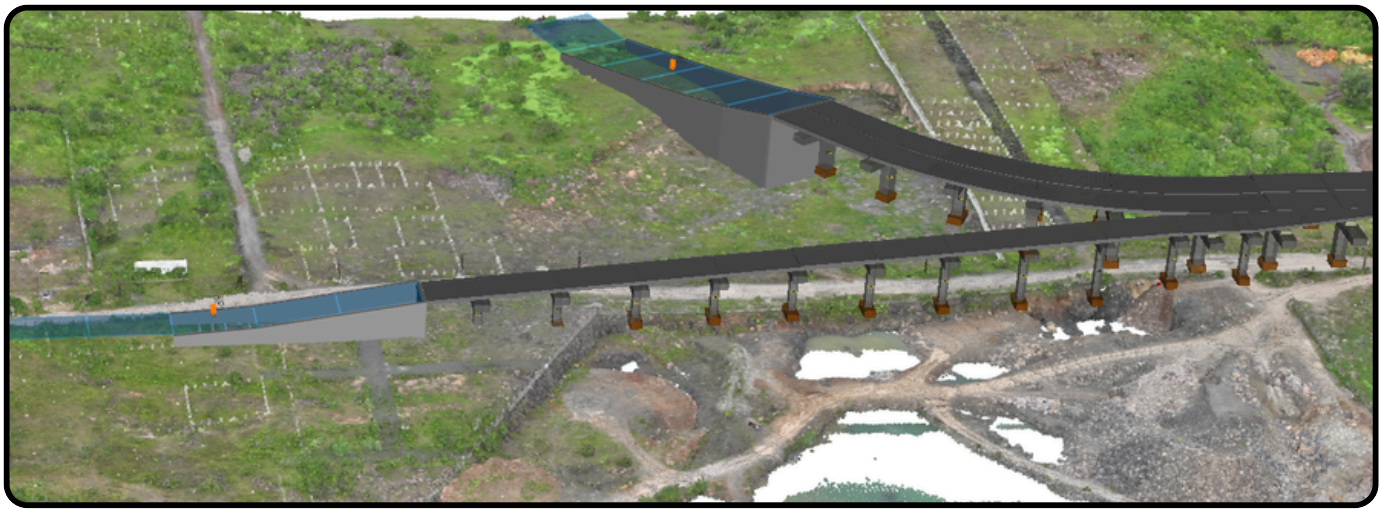
- Merge multiple design layers into a single unified model.
- Supports **Road–Road, Road–Bridge, and Bridge–Bridge** design merging.
- Identify and manage intersection points between different infrastructure elements.
- Allows users to import and merge designs from multiple sources into one project.
- Perform unified analysis for alignment, continuity, and material estimation.
- Improves coordination, design accuracy, and overall project efficiency.



Road & Bridge Merger



Bridge & Bridge Merger



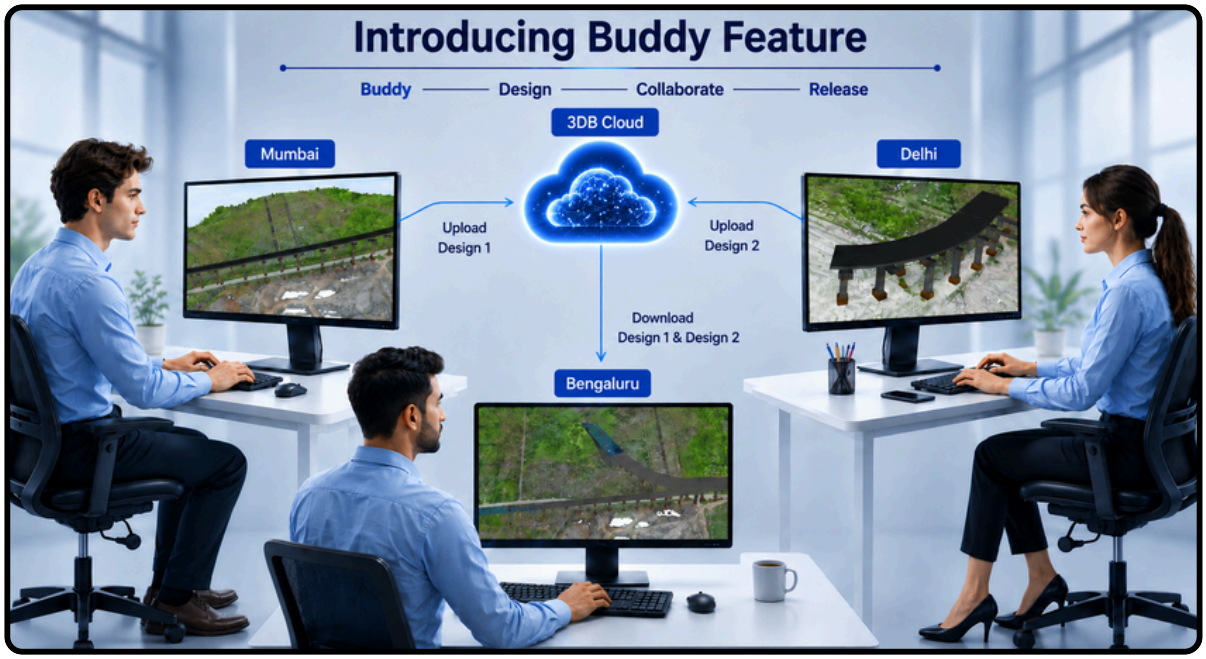
Merger & Intersection Point Design

4. Buddies Module

The Buddies Module is a collaborative feature within 3D Bharat that allows users to **connect with others** (buddies), **share designs**, and work together on infrastructure projects. It creates a network where users can **exchange ideas, improve designs**, and build projects collectively.

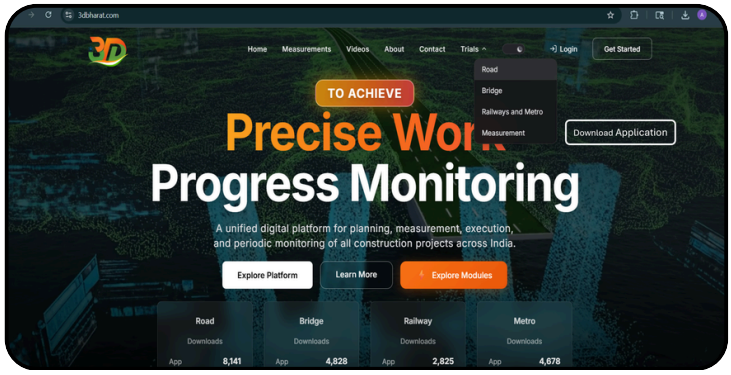
Buddy Features:

- Buddy users can share and merge their designs to create a complete and holistic project scenario.
- Supports design integration for interchange and intersection points.
- Enables merging of **Road to Bridge** connections.
- Supports **Road to Road alignment and junction design.**
- Allows **Bridge to Bridge** integration for **multi-section traffic bridge designs**.
- Supports incorporation of **side roads and connecting access roads.**
- Improves collaborative workflow for complete infrastructure planning and analysis.



4.1 How to Use the Buddies Module (Step-by-Step):

1. Download the Application
2. Visit the official website www.3dbharat.com and go to the trial section. Download the required application such as Road, Bridge, Railway, or Metro.



4.2 Add a Buddy

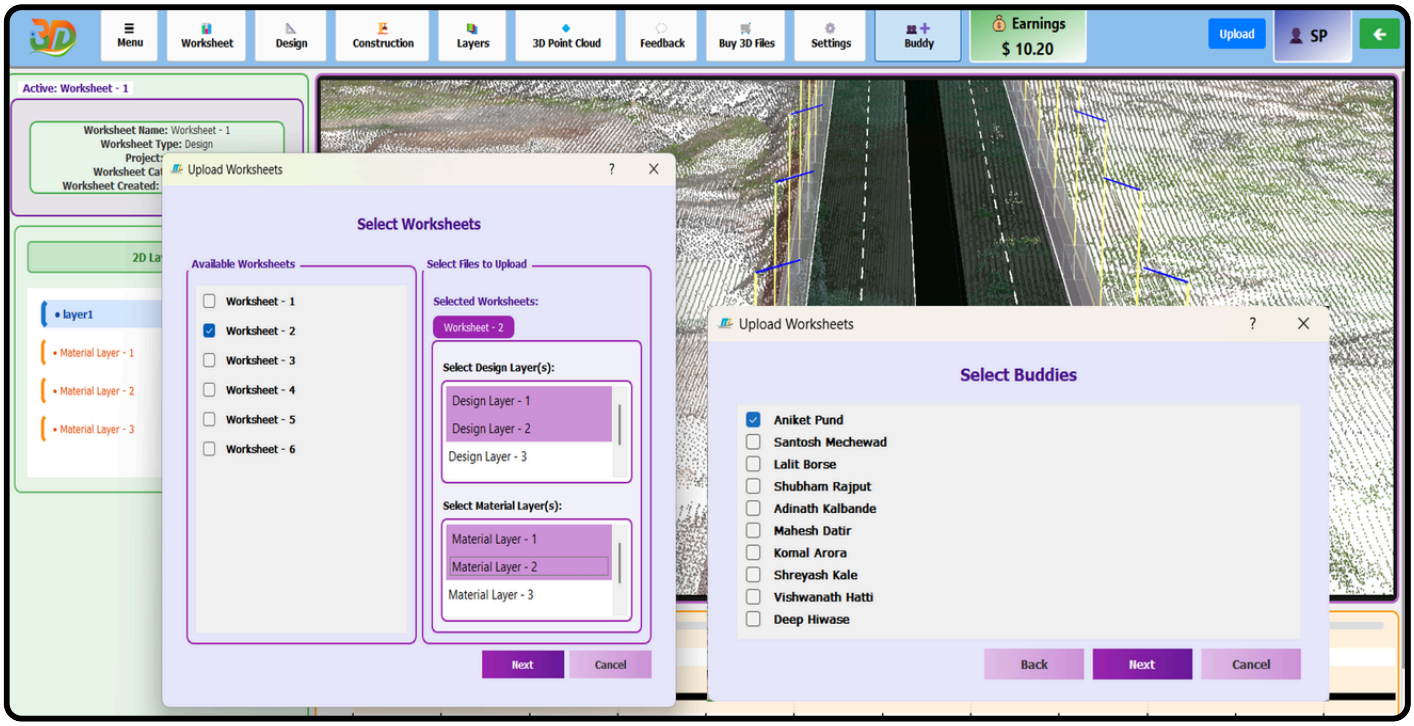
Open the Buddies Module and add your friend by entering their contact details.

- If the user is not registered, they will receive an **email invitation** to join **3D Bharat**.
- If already registered, they will be directly added as your buddy.



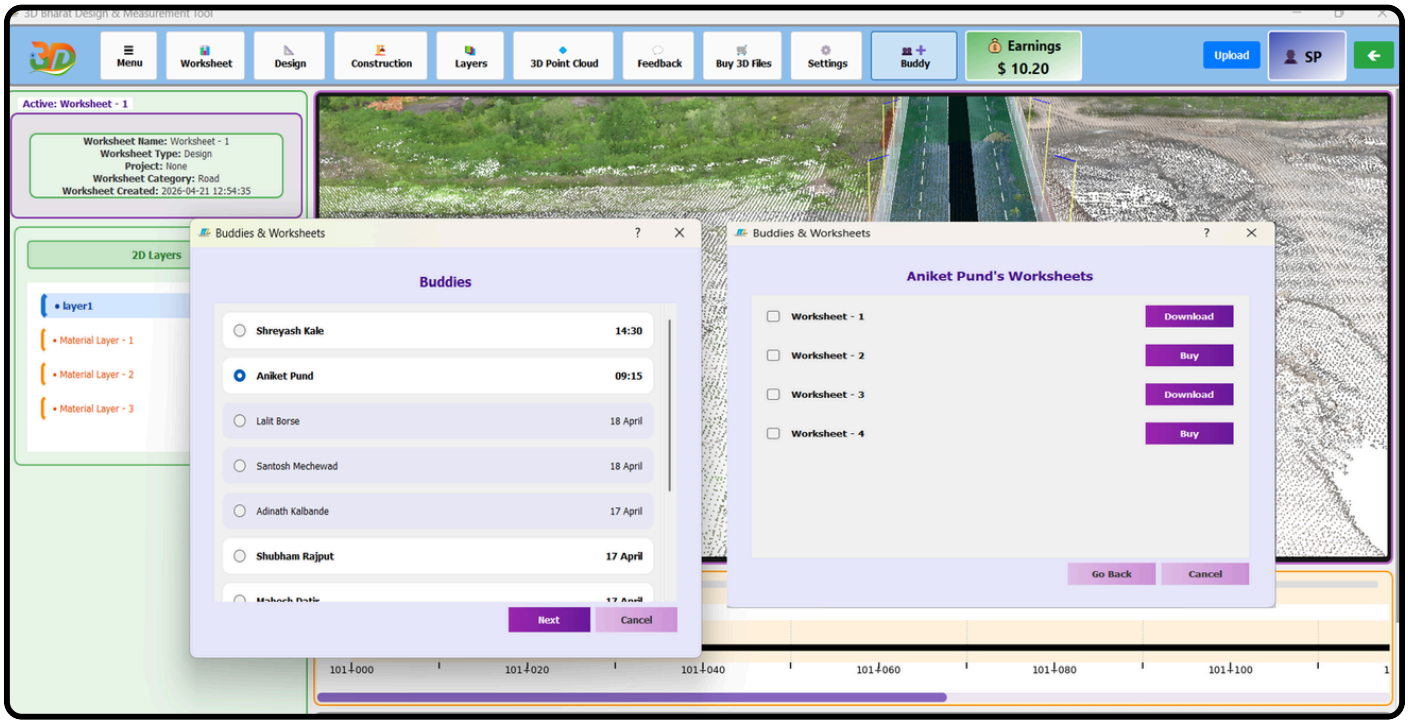
4.3 Share the Design

Once connected, you can share your design with your buddy through the platform.



4.4 Download & Collaborate

Your buddy can download the shared design, visualize it, analyze it & make modifications or merge it with other designs.



7. 3D Bharat Web Visualization Module

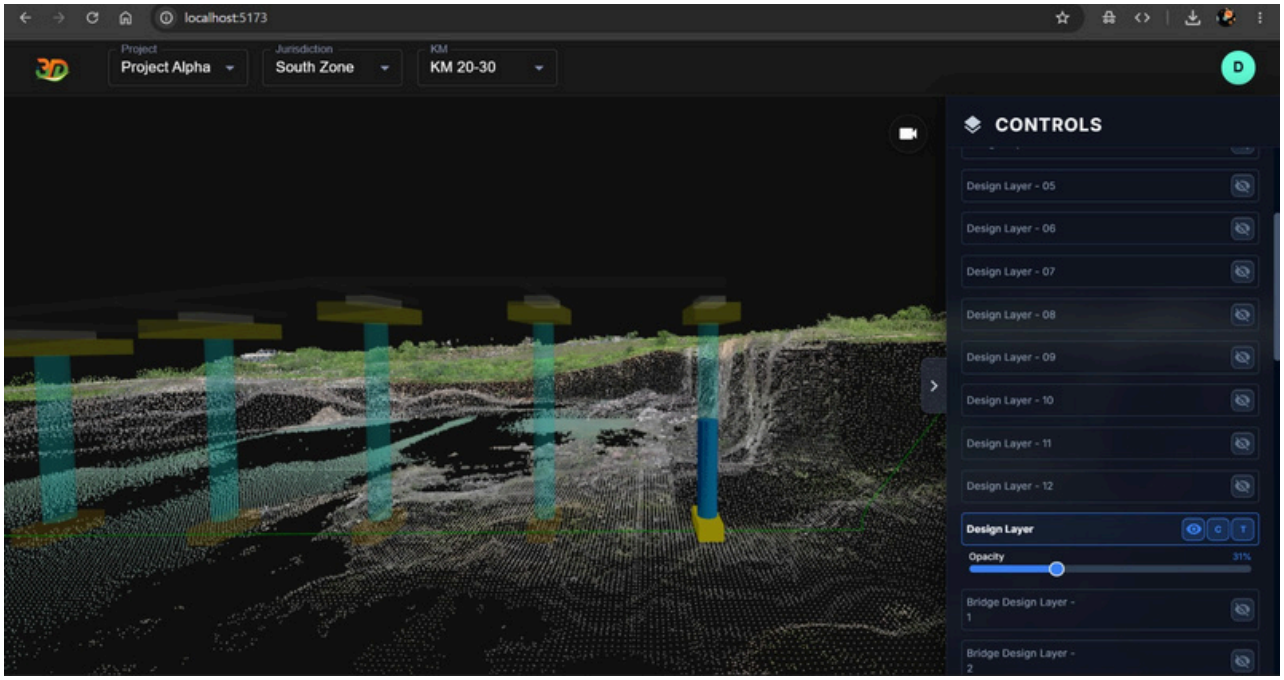
The 3D Bharat Web Visualization Module enables users to visualize 3D Point Cloud data integrated with **infrastructure designs directly in a web browser** for monitoring and analysis.

Capabilities:

- View 3D Point Cloud data with **infrastructure designs**.
- Visualize roads, bridges, metro, tunnels, cable-stayed bridges, and solar panel layouts.
- Compare design data with ground surveys and drone survey data.
- Monitor **construction progress in real time.**
- Visualize complex designs using JSON data.
- Perform design **validation and change detection.**
- Map project assets with geospatial references.
- **Access projects from anywhere** through a web browser.
- Enable collaboration between **departments and contractors.**
- Support **Digital Twin–based** project monitoring.



Interchanging Point on Web



Compare Design vs Actual Ground Data

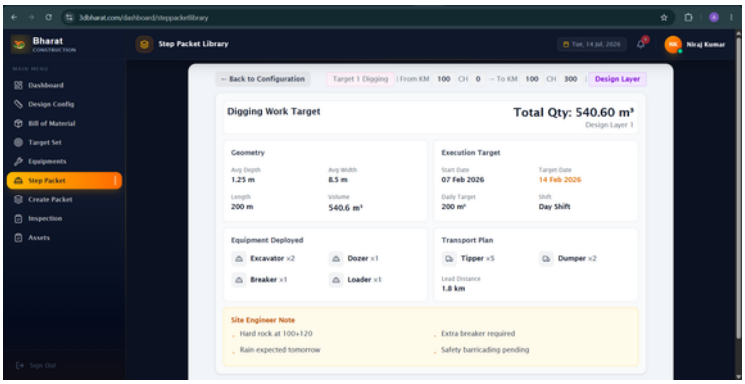
5. 3D Bharat Dashboard

The 3D Bharat Dashboard is the central project management module where departments and contractors configure, plan, and monitor all construction activities. It helps organize project execution by defining work steps, work packages, targets, and equipment requirements.

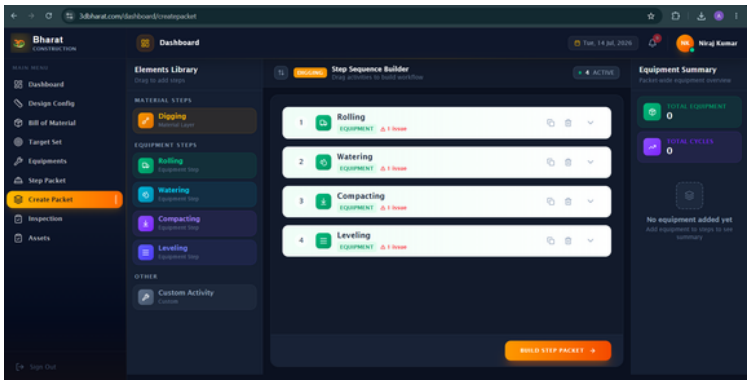
Features

- Create and manage construction projects.
- Define work steps such as **digging, excavation, material filling, rolling, watering, compaction, paving, and other construction activities.**
- Group multiple work steps into **work packages** for better planning and execution.
- **Create project targets** and milestones for each work package.
- Assign contractors, departments, and engineers to specific work packages.
- Configure the **equipment and machinery** required to complete each target.
- Configure project inspections for different construction stages and site activities.
- **Schedule inspections** and assign inspectors or responsible authorities.
- Record inspection observations, remarks, and approvals for quality assurance and compliance.
- Track the progress of work packages, project targets, and inspections in real time.
- Monitor **planned design vs. actual site progress.**
- Generate progress reports and performance analytics.
- Improve coordination between departments, contractors, inspectors, and field teams.
- Centralize project planning, monitoring, inspections, and reporting on a single platform.

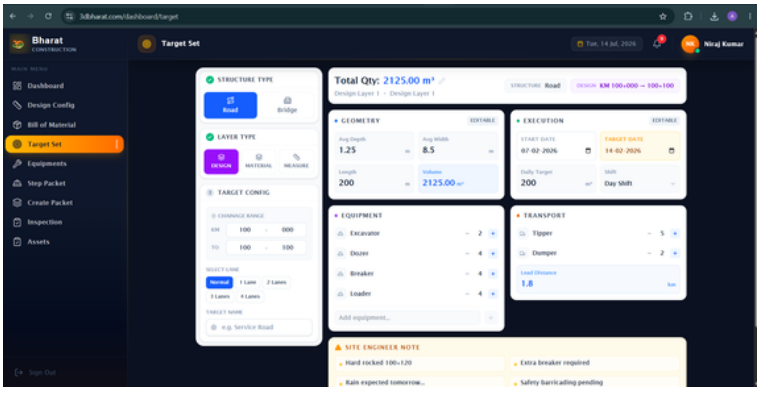
The 3D Bharat Dashboard provides a structured workflow for project planning, execution, inspection management, resource allocation, and real-time progress monitoring. It ensures efficient, transparent, and quality-driven management of construction projects from planning through completion



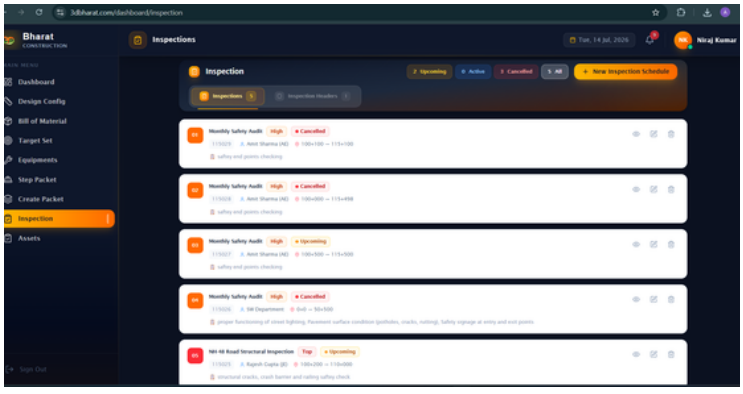
Steps Packet Configure



Create Packet



Target Setting



Inspection Configure

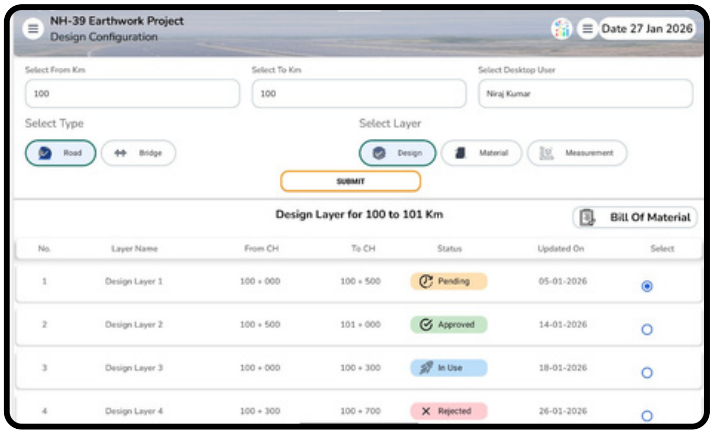
5. 3D Bharat Tablet Module

The 3D Bharat Tablet Module is designed for **field engineers, supervisors, and contractor employees** to capture and update construction data directly from the project site. It enables users to record ground progress against the approved design using KM, chainage, and block references, ensuring accurate and real-time project monitoring.

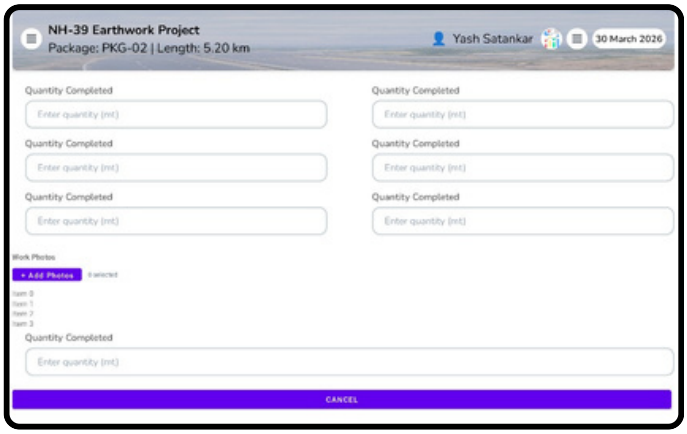
Features

- Record construction progress **block-wise, KM-wise, and chainage-wise.**
- Capture ground execution data directly from the project site using a tablet.
- View the approved **design, material quantities, and asset information** for the selected KM and chainage.
- View assigned **work targets, work packages, and work steps** based on the user's assigned tasks.
- Enter actual work progress against planned activities.
- Update the status of completed, ongoing, and pending work.
- Record **material consumption** and work **quantities.**
- View **inspection schedules, inspection remarks, inspection history and compliance requirements.**
- Capture site observations, comments, and progress notes.
- Upload **site photographs** and supporting **documents.**
- Record **GPS-based project location** for accurate field reporting.
- Synchronize field data with the central server.
- Enable real-time monitoring by departments and contractors.
- Reduce paperwork and improve the accuracy of field data collection.
- Maintain a complete digital record of construction activities.

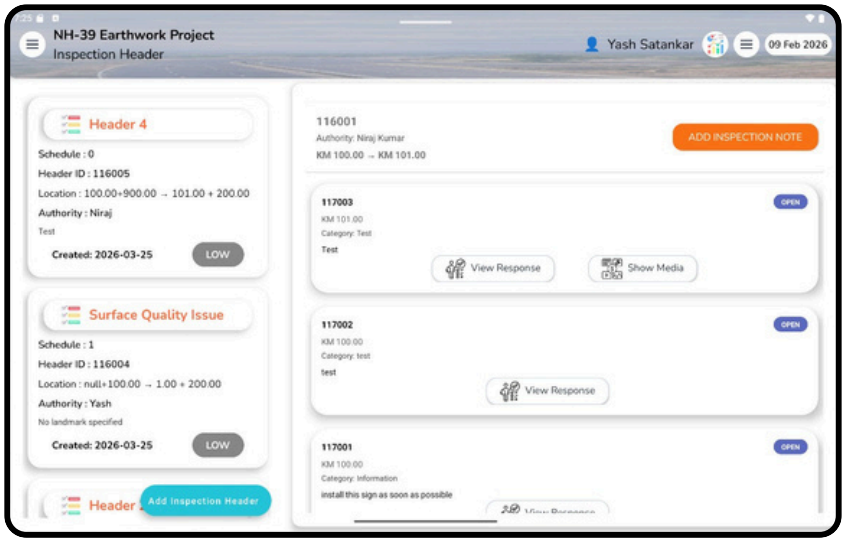
The 3D Bharat Tablet Module bridges the gap between design and field execution by enabling construction teams to collect, validate, and synchronize project data in real time, ensuring accurate progress monitoring and efficient project management.



Design Data



Daily Work Progress



Inspections Report